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UNIT 10

OSCILLATIONS

Life is a constant oscillation between the sharp horns of a dilemma – H.L. Mencken

Learning Objectives

In this unit, the student is exposed to

- oscillatory motion – periodic motion and non-periodic motion
- simple harmonic motion
- angular harmonic motion
- linear harmonic oscillator – both horizontal and vertical
- combination of springs – series and parallel
- simple pendulum
- expression of energy – potential energy, kinetic energy and total energy
- graphical representation of simple harmonic motion
- types of oscillation – free, damped, maintained and forced oscillations
- concept of resonance

BRTMPX

10.1 INTRODUCTION



Figure 10.1. Thanjavur dancing doll

Have you seen the Thanjavur Dancing Doll (In Tamil, it is called 'Thanjavur thalayatti bommai')?. It is a world famous Indian cultural doll (Figure 10.1). What does this doll do when disturbed? It will dance such that the head and body move continuously in a to and fro motion, until the movement gradually stops. Similarly, when we walk on the road, our hands and legs will move front and back. Again similarly, when a mother swings a cradle to make her child sleep, the cradle is made to move in to and fro motion. All these motions are different from the motion that we have discussed so far. These motions are shown in Figure 10.2. Generally, they are known as oscillatory motion or vibratory motion. A similar motion occurs even at atomic levels. When the temperature is raised, the atoms in a solid vibrate about their rest position (mean position or equilibrium position). The study of vibrational motion is very important in engineering applications, such as, designing the structure of building, mechanical equipments, etc.